

Marco Ciccone

PERSONAL INFORMATION

Surname: Ciccone
Name: Marco
Nationality: Italian
Address: Dept. of Electronics Information and Bioengineering
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Politecnico di Milano

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RESEARCH INTERESTS

Deep Learning, Meta Learning, Transfer Learning, Continual Learning,
Game Theory, Unsupervised Learning, Optimization, Computer Vision

EDUCATION

Politecnico di Milano, Italy

PhD candidate in Computer Science and Engineering

4th year

“Processing and Learning Dynamics of Deep Neural Networks”

Advisor: Prof. Matteo Matteucci

Co-Advisor: Phd Jonathan Masci

I am both interested in applied and theoretically grounded research.

I have mostly worked on deep learning models for computer vision problems.

My research is currently focused on:

- Meta-learning: fast adaptation and skills decomposition
- Representation learning: invariance properties and adversarial robustness

I am also interested in:

- Game Theory and Multi-agents systems
- Information Geometry

Romanian Institute of Science and Technology, Cluj-Napoca, Romania

Visiting PhD student, advised by PhD Luigi Malago’

“Information Geometry of Adversarial Robustness of Neural Networks”

Politecnico di Milano, Italy

MSc in Computer Science and Engineering

April 2016

- Thesis: *“Pixelwise Semantic Segmentation of urban scenes using Recurrent Deep Neural Networks”*
- Advisors: Prof. Matteo Matteucci, Phd Francesco Visin
- Language: English
- Grade: *110/110 cum Laude*
- GPA: *28.85/30*

- Thesis: “*Blind Source Separation methods for multichannel audio signals based on Independent Component Analysis*”
- Advisor: Prof. Fabrizio Argenti
- Language: Italian
- Grade: 97/110

PUBLICATIONS

- [1] M. Cannici, **M. Ciccone**, A. Romanoni, and M. Matteucci, “A differentiable recurrent surface for asynchronous event-based data,” in *European Conference on Computer Vision*, pp. 1–17, Springer, 2020
- [2] M. Cannici, **M. Ciccone**, A. Romanoni, and M. Matteucci, “Asynchronous convolutional networks for object detection in neuromorphic cameras,” *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, 2019
Best paper award, Second International Workshop on Event-based Vision and Smart Cameras Workshop at CVPR2019
- [3] M. Cannici, **M. Ciccone**, A. Romanoni, and M. Matteucci, “Attention mechanisms for object recognition with event-based cameras,” *Winter Conference on Applications of Computer Vision (WACV)*, 2019
- [4] **M. Ciccone**, M. Gallieri, J. Masci, C. Osendorfer, and F. Gomez, “Nais-net: Stable deep networks from non-autonomous differential equations,” in *Advances in Neural Information Processing Systems 32 (NeurIPS)*, 2018
- [5] F. Lattari*, **M. Ciccone***, M. Matteucci, J. Masci, and F. Visin, “Reconvnet: Video object segmentation with spatio-temporal features modulation,” *The 2018 DAVIS Challenge on Video Object Segmentation - CVPR Workshops*, 2018
- [6] A. Romanoni, **M. Ciccone**, F. Visin, and M. Matteucci, “Multi-view stereo with single-view semantic mesh refinement,” in *IEEE International Conference on Computer Vision (ICCV)*, pp. 706–715, 2017
- [7] F. Visin, **M. Ciccone**, A. Romero, K. Kastner, C. Kyunghyun, Y. Bengio, M. Matteucci, and A. Courville, “ReSeg : A Recurrent Neural Network-based Model for Semantic Segmentation,” in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, 2016
Best paper award, DeepVision Workshop at CVPR2016

ACADEMIC
EXPERIENCE

Deep Learning 2017/2018, Prof. Matteo Matteucci

PhD and MSc Course at Politecnico di Milano

Teaching Assistant

February 2018

- Tensorflow and Pytorch Tutorial
- Advanced Convolutional Neural Networks

Cognitive Robotics 2016/2017, Prof. Matteo Matteucci

MSc Course at Politecnico di Milano

Teaching Assistant

June 2017

- Deep Learning Introduction
- From Linear Models to Deep Neural Networks
- Convolutional and Recurrent Neural Networks

CONFERENCES AND COURSES	Transylvanian Machine Learning Summer School 2018 , Cluj, Romania <i>Acceptance Rate: 8%</i> 16-22 July 2018
	Machine Learning Summer School 2017 , Tübingen, Germany <i>Acceptance Rate: 13%</i> 19-30 June 2017
	International Computer Vision Summer School 2017 , Sicily, Italy <i>Acceptance Rate: 35%</i> 9-15 July 2017
	International Conference on Acoustics, Speech, and Signal Processing ICASSP 2014 , Florence, Italy <i>Staff volunteer</i> 4-9 May 2014 Attended tutorials : - Informed Audio Source Separation: Trends, Approaches and Algorithms - Deep learning for natural language processing and related applications
PROFESSIONAL EXPERIENCE	NVIDIA , Santa Clara, California <i>Research Intern</i> August 2018 - November 2018 Project on adversarial robustness of Neural Networks.
	NNAISENSE , Lugano, Switzerland <i>Research Intern</i> July 2017 - January 2018 I spent 7 months as research intern at NNAISENSE tutored by Jonathan Masci and Marco Gallieri. In this time I worked on stability and robustness of Residual Networks from a control theory perspective.
	Horus Technology , Milan, Italy <i>Machine Learning Engineer</i> June 2016 - June 2017 Horus Technology is an Italian start-up (recently rebranded as Eyra) that operates in the field of Artificial Intelligence. Its first product, Horus, is a wearable device that aims to assist blind and visually impaired people thanks to state-of-the-art computer vision technology. During my time here, I worked on deep learning models for Object Recognition and Scene Analysis to be deployed on a low power embedded system based on NVIDIA TK1.
	Projest S.p.A. , Prato, Italy <i>IT Consultant - Software Engineer</i> Nov 2009 - Oct 2012 I carried out several consulting projects as IT consultant for Guccio Gucci S.p.A., Guccio Logistica S.p.A. and other clients.
LANGUAGES AND CERTIFICATES	English: Highly proficient in spoken and written English Italian: Native speaker French: Basic communication skills